EXPERIMENT - 18

18. Implement image cropping, copying and pasting to select a region of interest (ROI) from the source image using Open CV.

Aim

To implement image cropping, copying, and pasting to select a Region of Interest (ROI) from the source image using OpenCV.

Procedure

1. Import the OpenCV library.
2. Read the input image.
3. Select a Region of Interest (ROI) using pixel coordinates.
4. Crop the selected region from the original image.
5. Copy the cropped region.
6. Paste the copied region into another location in the image.
7. Display the original image, cropped image, and final output image.

Program

```
import cv2

img = cv2.imread("image.jpg")

if img is None:
    print("Image not found")
else:
    img = cv2.resize(img, (500, 350))

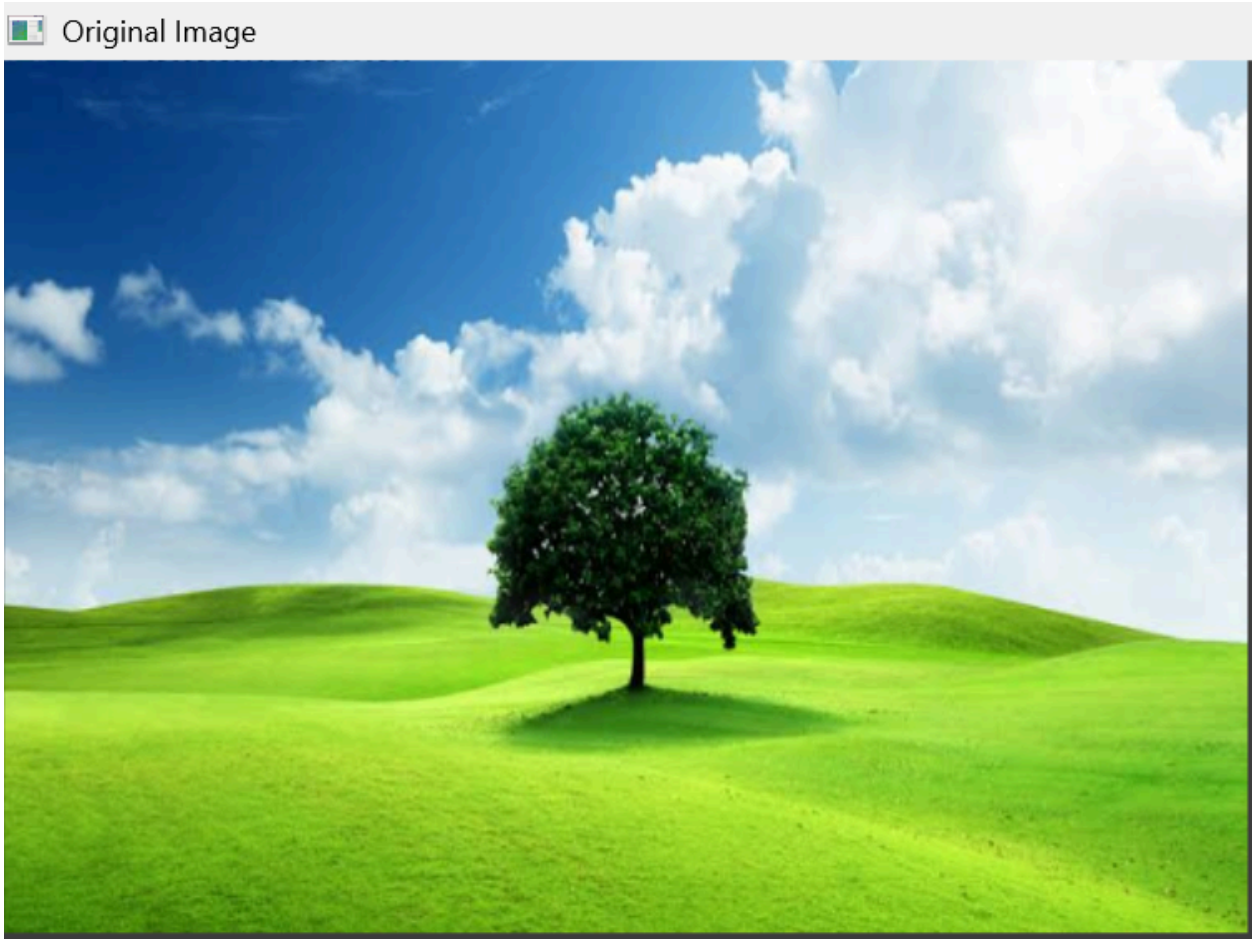
    roi = img[50:150, 50:150]
    copied = roi.copy()

    output = img.copy()
    output[180:280, 300:400] = copied
```

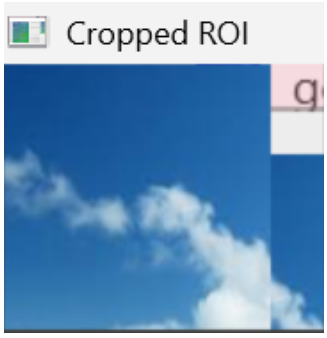
```
cv2.imshow("Original Image", img)
cv2.imshow("Cropped ROI", roi)
cv2.imshow("Copy and Paste Output", output)

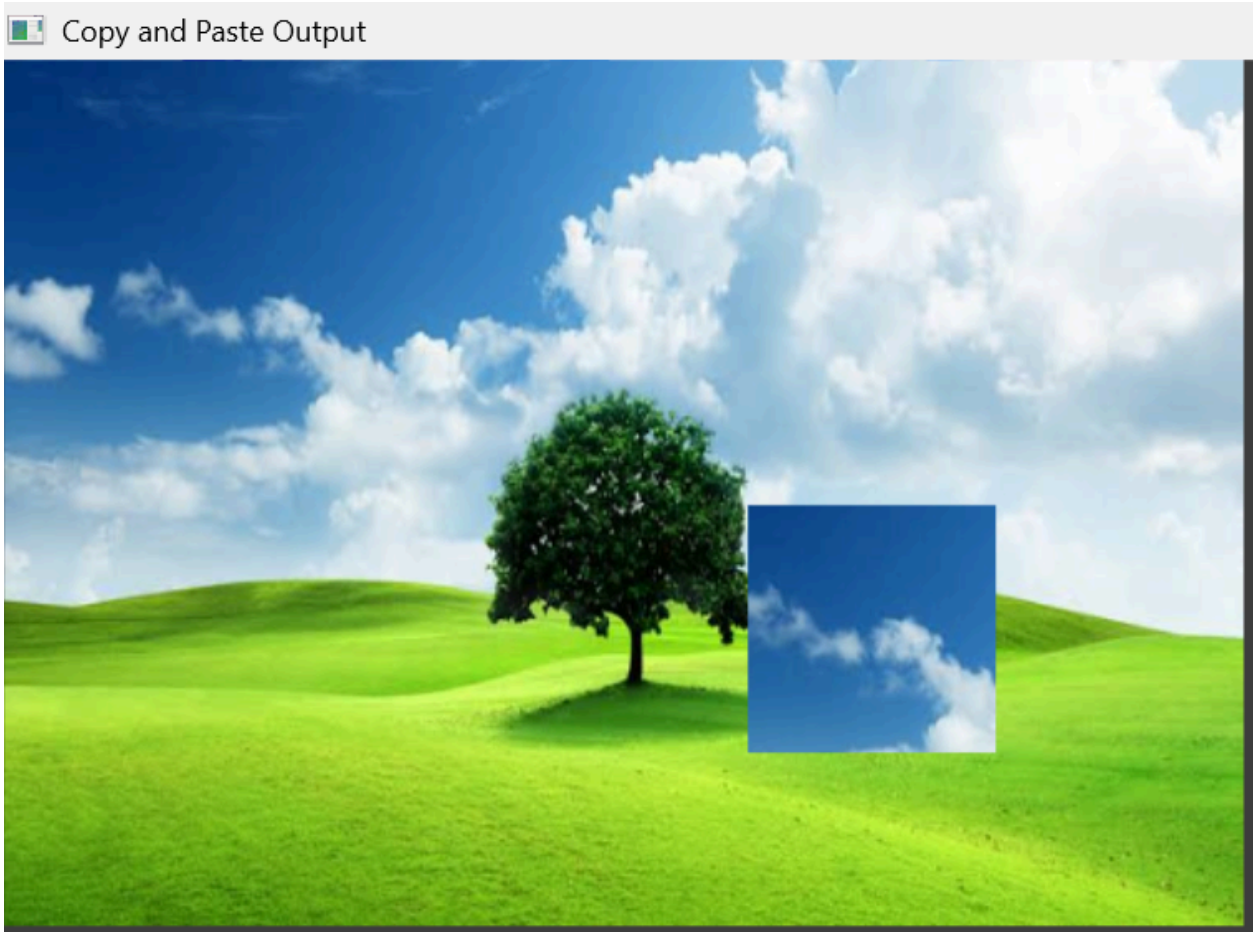
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Sample Input



Sample Output





Result

Image cropping, copying, and pasting were successfully implemented using OpenCV to select and reposition a Region of Interest (ROI).